

MAT344 Notes

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1 Differentiability in $\mathbb{C}$

1.0 Precalculus review

Definition 1.1 ($\mathbb{C}$) — The *complex numbers* $\mathbb{C}$ is the direct sum $\mathbb{R} \oplus i\mathbb{R}$, where $i \equiv (0, 1)$, $1 \equiv (1, 0)$; it is given the structure of a commutative, associative, unital $\mathbb{R}$ -algebra with the multiplication

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc)$$

(so in particular $i^2 = -1$).

Exercise 1.2. A list of properties you should remember from high school:

- (a) Show that $\mathbb{C}$ is a field, in with $(a + bi)^{-1} = \frac{a-bi}{a^2+b^2}$.
- (b) Show that complex conjugation, defined as $\overline{a + bi} \equiv a - bi$, is a field automorphism of $\mathbb{C}$ that fixes $\mathbb{R}$.
- (c) Show that the map $|\bullet| : \mathbb{C} \rightarrow \mathbb{R}_+$ given by $|a + bi| = \sqrt{a^2 + b^2}$ (equivalently, $|z|^2 = z\bar{z}$) is a norm on $\mathbb{C}$.
- (d) Show that any $z \in \mathbb{C} - \{0\}$ can be written uniquely as $re^{i\theta} = r(\cos \theta + i \sin \theta)$ for $r \in \mathbb{R}_+$ and $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ (here, the exponential map is defined via its power series). In particular, this suggests that multiplication by $z = re^{i\theta}$ corresponds to scaling about the origin by r and rotating counterclockwise by θ .

Property (d) in particular suggests why $\mathbb{C}$ is so nice: multiplication by a complex number is much more rigid than any old linear transformation on $\mathbb{R}^2$, as we will see shortly.

1.1 Holomorphic functions

Recall that a map $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable if it locally behaves like an $\mathbb{R}^n$ -linear map. Naturally, one gets the following definition for complex differentiability:

Definition 1.3 (Holomorphic) — A function $f: U \rightarrow \mathbb{C}$ defined on an open subset $U \subset \mathbb{C}$ is *holomorphic* or *complex analytic* at $z_0 \in U$ if

$$f'(z_0) \equiv \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists in $\mathbb{C}$ (i.e. the limit is taken over $h \in \mathbb{C}$ tending to 0).

As with functions on $\mathbb{R}$, sums, products, quotients (where defined), and compositions (where defined) of holomorphic functions are also holomorphic.

1.1.1 Conformal linear maps, Cauchy-Riemann equations

Observe that a $\mathbb{C}$ -linear map $L: \mathbb{C} \rightarrow \mathbb{C}$ is uniquely determined by $L(1)$, since $L(z) = zL(1)$ by linearity; thus any $\mathbb{C}$ -linear map corresponds to multiplying by a fixed complex number $a + bi$. As an $\mathbb{R}$ -linear map, multiplication by $a + bi$ can be expressed as the matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

with respect to the standard basis (1 and i) of $\mathbb{C}$.

As mentioned at the end of section 1.0, such linear transformations are angle- and orientation-preserving (equivalently, they preserve directed angles — that is, if ℓ_1 and ℓ_2 are lines such that the counterclockwise angle from ℓ_1 to ℓ_2 is θ , then the counterclockwise angle from $L(\ell_1)$ to $L(\ell_2)$ is also θ). Such a linear map is called *conformal*.

Remark 1.4. More generally, we call a differentiable map $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (or between two Riemannian manifolds of dimension 2) conformal if it preserves angles between curves, where the angle between two curves at a given point of intersection p is the angle between the tangent lines at that point; this is equivalent to df_p being a conformal linear map.

Proposition 1.5. All conformal $\mathbb{R}$ -linear maps $L: \mathbb{C} \rightarrow \mathbb{C}$ are $\mathbb{C}$ -linear.

Proof. Let $z \in \mathbb{C}$. The product iz is z rotated 90° counterclockwise about the origin, so $L(iz)$ is the counterclockwise 90° rotation of $L(z)$ about the origin, i.e. $L(iz) = iL(z)$. Thus, by $\mathbb{R}$ -linearity,

$$L((a + bi)z) = aL(z) + bL(iz) = (a + bi)L(z)$$

for any $a, b \in \mathbb{R}$, which means L is $\mathbb{C}$ -linear. $\square$

Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be a holomorphic function. We can consider f as a map on $\mathbb{R}^2$ by treating it as a function of the real and imaginary parts; in this context, the derivative of $f(x, y) = (f_1(x, y), f_2(x, y))$ (where $f_1, f_2: \mathbb{R}^2 \rightarrow \mathbb{R}$) is the linear transformation

$$df = \begin{pmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \end{pmatrix}.$$

The limit defining the derivative of f corresponds to this linear transformation, so df must be a conformal linear map; this means that we must have

$$\frac{\partial f_1}{\partial x} = \frac{\partial f_2}{\partial y}, \quad \frac{\partial f_1}{\partial y} = -\frac{\partial f_2}{\partial x}$$

at any point at which f is holomorphic. This system of equations is called the *Cauchy-Riemann equations*. In fact, a partial converse holds too: the Cauchy-Riemann equations exactly determine whether f is complex differentiable at a given point if f is sufficiently regular.

Proposition 1.6. Let $f: U \rightarrow \mathbb{C}$, with $U \subseteq \mathbb{C}$ an open subset, have continuous partial derivatives on all of U (we say that such a function f is C^1 , since it is C^1 as a function on (a subset of) $\mathbb{R}^2$). Then, f is holomorphic if and only if the Cauchy-Riemann equations hold for all $z_0 \in U$.

Proof. If f is holomorphic, then the limits

$$\begin{aligned} \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{f(z_0 + h) - f(z_0)}{h} &= \frac{\partial f}{\partial x}(z_0) = \frac{\partial f_1}{\partial x}(z_0) + i \frac{\partial f_2}{\partial x}(z_0) \\ \lim_{\substack{h \rightarrow 0 \\ h \in i\mathbb{R}}} \frac{f(z_0 + h) - f(z_0)}{h} &= \frac{1}{i} \frac{\partial f}{\partial y}(z_0) = -i \frac{\partial f_1}{\partial y}(z_0) + \frac{\partial f_2}{\partial y}(z_0) \end{aligned}$$

are equal. Comparing the real and imaginary parts yields the Cauchy-Riemann equations.

Conversely, if f has continuous partial derivatives, then it is differentiable as a map on (a subset of) $\mathbb{R}^2$. This derivative obeys the Cauchy-Riemann equations, so it is a $\mathbb{C}$ -linear map, say corresponding to multiplication by $z \in \mathbb{C}$. Thus, by definition of differentiability in $\mathbb{R}^n$,

$$0 = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0) - df_{z_0}h}{h} = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0) - zh}{h}$$

so the limit defining $f'(z_0)$ exists. $\square$

Remark 1.7 (Looman-Menchoff theorem). In fact, any continuous function $f: U \rightarrow \mathbb{C}$ whose partials exist and satisfy the Cauchy-Riemann equations on all of U is holomorphic on U ; this is called the *Looman-Menchoff theorem*. The continuity assumption is required because $f(z) = \exp(-z^{-4})$ has partials that exist and satisfy Cauchy-Riemann everywhere, but is not continuous (hence not

holomorphic) at zero. Additionally, if f is only required to have partials satisfying the Cauchy-Riemann equations at a single point, then $f: \mathbb{C} \rightarrow \mathbb{C}$ defined as $f(0) = 0$ and $f(z) = z^5/|z|^4$ for $z \neq 0$ is continuous at 0 but not holomorphic at 0, since

$$\lim_{z \rightarrow 0} \frac{f(z) - 0}{z} = \lim_{z \rightarrow 0} \frac{z^4}{|z|^4}$$

is easily seen to not exist, as taking the limit along the real line and taking the limit along the line $\operatorname{Re} z = \operatorname{Im} z$ respectively give 1 and -1 .

1.1.2 The operators $\frac{\partial}{\partial z}$ and $\frac{\partial}{\partial \bar{z}}$

Definition 1.8 ($\frac{\partial}{\partial z}$ and $\frac{\partial}{\partial \bar{z}}$) — The formal operators $\frac{\partial}{\partial z}$ and $\frac{\partial}{\partial \bar{z}}$ are defined as

$$\frac{\partial}{\partial z} \equiv \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} \equiv \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

Proposition 1.9. A C^1 function $f: U \rightarrow \mathbb{C}$ (where, as usual, $U \subseteq \mathbb{C}$ is open) is holomorphic if and only if $\frac{\partial}{\partial \bar{z}} f = 0$ on all of U .

Proof. Write $f = f_1 + if_2$ for real-valued functions f_1 and f_2 . We can manually compute

$$2 \frac{\partial}{\partial \bar{z}} f = \frac{\partial f_1}{\partial x} - \frac{\partial f_2}{\partial y} + i \left(\frac{\partial f_1}{\partial y} + \frac{\partial f_2}{\partial x} \right),$$

which is zero if and only if the real and imaginary parts are zero, which is equivalent to the Cauchy-Riemann equations. The statement then follows from Proposition 1.6. $\square$

Remark 1.10. We can compute

$$\frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \frac{1}{4} \left(\frac{\partial^2}{\partial^2 x} + \frac{\partial^2}{\partial^2 y} \right) = \frac{1}{4} \Delta$$

where $\Delta = \frac{\partial^2}{\partial^2 x} + \frac{\partial^2}{\partial^2 y}$ is called the *Laplacian* of f .

Proposition 1.11. Let $f: U \rightarrow \mathbb{C}$ be holomorphic at $z_0 \in U$. Then

$$\frac{\partial}{\partial z} f(z_0) = f'(z_0).$$

Proof. Write $f = f_1 + if_2$ for real-valued functions $f_1, f_2: U \rightarrow \mathbb{R}$. Then,

$$\frac{\partial}{\partial z} f = \frac{1}{2} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} \right) + \frac{i}{2} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right).$$

Since f is holomorphic at z_0 , the Cauchy-Riemann equations imply that multiplication by $\frac{\partial}{\partial z} f(z_0)$ corresponds to the linear transformation

$$\begin{pmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \end{pmatrix}$$

1.2 Functions on Riemann surfaces

Let $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$ be the one-point compactification of $\mathbb{C}$ (which is homeomorphic to S^2). Let $U = \mathbb{C}_\infty - \{\infty\} = \mathbb{C}$ and $V = \mathbb{C}_\infty - \{0\}$, and define

$$\begin{aligned}\varphi: U &\rightarrow \mathbb{C}, & f(z) &= z, \\ \psi: V &\rightarrow \mathbb{C}, & f(z) &= \frac{1}{z}\end{aligned}$$

with the convention $\frac{1}{\infty} = 0$. These are both homeomorphisms (since they are continuous and are both their own inverse) and the transition map

$$\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V), \quad \psi \circ \varphi^{-1}(z) = \frac{1}{z}$$

is a *biholomorphism* (holomorphic map with holomorphic inverse), so the charts (U, φ) and (V, ψ) turn $\mathbb{C}_\infty$ into a *complex 1-manifold*, also called a *Riemann surface*. The Riemann surface $\mathbb{C}_\infty$ is known as the *Riemann sphere*.

Concretely, the idea of the maps φ and ψ is that they describe how $\mathbb{C}_\infty$ looks like some copies of $\mathbb{C}$ glued together, with $\varphi \circ \varphi^{-1}$ being a biholomorphism meaning that the way they are glued is compatible with the kind of structure on $\mathbb{C}$ that we care about. Since being holomorphic only depends on the behavior of a function locally, we can use these charts to define holomorphic functions between any two Riemann surfaces.

Example 1.12 (Automorphisms of the Riemann sphere). Let $a, b, c \in \mathbb{C}$ with $a \neq b$ and $c \neq 0$. The function $f: \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ defined by $f(z) = c \cdot \frac{z-a}{z-b}$ when $z \notin \{b, \infty\}$, $f(b) = \infty$, and $f(\infty) = c$ is a biholomorphism of $\mathbb{C}_\infty$; in fact, these are the only biholomorphisms on $\mathbb{C}_\infty$.

Example 1.13. Let $p, q: \mathbb{C} \rightarrow \mathbb{C}_\infty$ be complex polynomials with no common factors. Define

$$f(z) = \begin{cases} p(z)/q(z) & q(z) \neq 0 \\ \infty & q(z) = 0. \end{cases}$$

Since f is holomorphic whenever $q(z) \neq 0$ and $\frac{1}{f}$ is holomorphic whenever $p(z) \neq 0$, this makes $f: \mathbb{C} \rightarrow \mathbb{C}_\infty$ a holomorphic function. If $a \in \mathbb{C}$ is a point at which $f(z) = \infty$, there exists a positive integer n such that

$$f(z) = \frac{1}{(z-a)^n} g(z)$$

for holomorphic g that is nonzero on (a neighborhood of) a . The complex number a is called a *pole* of f , and the integer n is called its *order*. Similarly, if $f(b) = 0$, then $f(z) = (z-b)^n h(z)$ for h that is nonzero and holomorphic in a neighborhood of b ; in this case b is a *zero* of f with *order* n . Note that a pole of f with order n is a zero of $\frac{1}{f}$ with order n , and vice versa.

2 Differential geometry crash course

Definition 2.1 (Tangent vector, cotangent vector) — todo

Definition 2.2 (1-form) — todo

Definition 2.3 (Integral of 1-form) — todo

3 The Cauchy integral formula and consequences

write down important results from HWs

restructure sections

Now that we have established the basic geometric and algebraic terminology and some facts about $\mathbb{C}$ we can actually do some complex analysis.

3.1 The Cauchy integral formula

Theorem 3.1 (Cauchy's integral theorem). Let $\omega = f(z)dz$ be a holomorphic 1-form on a good domain Ω . Then

$$\int_{\partial\Omega} \omega = 0.$$

Proof. Apply Green's theorem to get

$$\int_{\partial\Omega} \omega = \int_{\Omega} d\omega = 0$$

where the integral is zero because $d\omega = 0$, as f is holomorphic. $\square$

Theorem 3.2 (Cauchy's integral formula). Let Ω be a good domain and $f \in \mathcal{O}(\Omega)$. For every $z \in \Omega$,

$$f(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Proof. Fix $z \in \Omega$ and let $\varepsilon > 0$ be such that the closed ball $\overline{B}_\varepsilon(z)$ is contained in Ω . Since $\frac{f(\zeta)}{\zeta - z}$ is a holomorphic function of ζ on $\Omega - B_\varepsilon(z)$, Green's theorem shows that

$$\int_{\partial\Omega} \frac{f(\zeta)}{\zeta - z} d\zeta = \int_{|\zeta - z| = \varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Then, parametrizing the circle in the obvious way, we get

$$\int_{|\zeta - z| = \varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta = \int_0^{2\pi} \frac{f(z + \varepsilon e^{i\theta})}{\varepsilon e^{i\theta}} \cdot \varepsilon i e^{i\theta} d\theta = i \int_0^{2\pi} f(z + \varepsilon e^{i\theta}) d\theta.$$

The equation we got from Green's theorem tells us that this integral is independent of ε , so it is equal to its limit as ε goes to zero; moreover, we can compute that

$$\left| \int_0^{2\pi} f(z + \varepsilon e^{i\theta}) - f(z) d\theta \right| < \int_0^{2\pi} C \varepsilon d\theta = 2\pi C \varepsilon,$$

where C is a constant such that $|f(z + \varepsilon e^{i\theta}) - f(z)| < C |\varepsilon e^{i\theta}|$ for sufficiently small ε , which exists because f is differentiable (namely take any $C > |f'(z)|$). In summary,

$$\begin{aligned} \int_{\partial\Omega} \frac{f(\zeta)}{\zeta - z} d\zeta &= \int_{|\zeta - z| = \varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta \\ &= i \int_0^{2\pi} f(z + \varepsilon e^{i\theta}) d\theta = i \cdot \lim_{\varepsilon \rightarrow 0} \int_0^{2\pi} f(z + \varepsilon e^{i\theta}) d\theta \\ &= i \int_0^{2\pi} f(z) d\theta = 2\pi i f(z). \end{aligned}$$

$\square$

Theorem 3.3 (Holomorphic functions are analytic). Let Ω be a good domain and $f \in \mathcal{O}(\Omega)$. For every $z_0 \in \Omega$, f has a convergent power series in a ball around z_0 .

Proof. Let r be the distance from z_0 to $\partial\Omega$ and suppose that $z \in \overline{B}_{r_0}(z_0)$ for some $r_0 < r$. Then

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)}{(\zeta - z_0) - (z - z_0)} d\zeta \\ &= \frac{1}{2\pi i} \int_{\partial\Omega} \left(\frac{f(\zeta)}{\zeta - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^n \right) d\zeta \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \cdot (z - z_0)^n \right) \end{aligned}$$

where we can expand the geometric series because $|z - z_0| < |\zeta - z_0|$ by assumption and we can pull the sum out of the integral because the geometric series converges uniformly over $z \in \overline{B}_{r_0}(z_0)$ (say, by the Weierstrass M -test). This is thus a power series expansion for $f(z)$ over all $z \in B_r(z_0)$. $\square$

Proposition 3.4 (Analytic continuation). Let $U \subseteq \mathbb{C}$ be open and connected and suppose that $f \in \mathcal{O}(U)$.

- (1) f is infinitely differentiable.
- (2) The zeroes of f are isolated unless f is the zero function on U .
- (3) If f and g are holomorphic functions on U that agree on any open set, then they are the same function.

Proof. (1) This is immediate from Theorem 3.3.

- (2) Suppose $z_0 \in U$ satisfies $f(z_0) = 0$. Then, in a neighborhood of z_0 ,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

If there is a $k > 0$ such that $a_k \neq 0$, take the smallest such k , so that $f(z) = (z - z_0)^k g(z)$ for g defined in a neighborhood of z_0 by

$$g(z) = \sum_{n=0}^{\infty} a_{n+k} (z - z_0)^n.$$

Since $g(z_0) = \lim_{z \rightarrow z_0} g(z) \neq 0$, g , hence f , is nonzero in a neighborhood of z_0 , so z_0 is an isolated zero of f .

On the other hand, if all the a_n are zero, then f is zero in a neighborhood of z_0 . This implies that the set of non-isolated zeroes of f is open in U . At the same time, the set of non-isolated zeroes must be closed in U , because $f^{-1}(0)$ is closed and a closed set minus some isolated points is still closed. Connectedness of U then tells us that the set of non-isolated zeroes of f is either all of U (in which case f is the zero function) or empty (in which case all the zeroes of f are isolated).

- (3) If f and g agree on an open set, then $f - g$ has a non-isolated zero, so $f - g \equiv 0$ by (2). $\square$

3.2 Derivatives as integrals

The fact that holomorphic functions are analytic and that their coefficients can be computed via certain integrals gives us the following formula:

Corollary 3.5. If f is holomorphic on a good domain Ω , then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

Proof. Immediate from differentiating the power series expansion around z (recall that the derivative of a power series is obtained by termwise differentiation). $\square$

One reason that this formula is useful because it lets us easily estimate derivatives; for example:

Lemma 3.6. If $\{f_k\}_{k=1}^\infty$ is a Cauchy sequence in $\mathcal{O}(\overline{B}_r(z_0))$, then, for all $n \geq 1$, the sequence of n th derivatives $\{f_k^{(n)}\}_{k=1}^\infty$ is also Cauchy when restricted to $\overline{B}_{r_0}(z_0)$ for any $r_0 < r$ (where the space of holomorphic functions is endowed with the sup metric).

Proof. Let $k, \ell \in \mathbb{N}$ be such that $|f_k - f_\ell| < \varepsilon$ on $\partial B_r(z_0)$. Then by Corollary 3.5,

$$f_k^{(n)}(z) - f_\ell^{(n)}(z) = \frac{n!}{2\pi i} \int_{|\zeta - z_0|=r} \frac{f_k(\zeta) - f_\ell(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

for all $z \in B_r(z_0)$. Suppose now that $z \in B_{r_0}(z_0)$, so that $|\zeta - z| \geq r - r_0$ for ζ on $\partial B_r(z_0)$. We then get

$$\begin{aligned} |f_k^{(n)}(z) - f_\ell^{(n)}(z)| &= \frac{n!}{2\pi} \left| \int_{|\zeta - z_0|=r} \frac{f_k(\zeta) - f_\ell(\zeta)}{(\zeta - z)^{n+1}} dz \right| \\ &= \frac{n!}{2\pi} \left| \int_0^{2\pi} \frac{f_k(\zeta) - f_\ell(\zeta)}{(\zeta - z)^{n+1}} \cdot ire^{i\theta} d\theta \right| \\ &\leq \frac{n! \cdot r}{2\pi} \int_0^{2\pi} \left| \frac{f_k(\zeta) - f_\ell(\zeta)}{(\zeta - z)^{n+1}} \right| d\theta \\ &< \frac{n! \cdot r}{2\pi} \int_0^{2\pi} \frac{\varepsilon}{(r - r_0)^{n+1}} d\theta = \frac{n! \cdot r \varepsilon}{(r - r_0)^{n+1}}. \end{aligned}$$

Since the coefficient of ε depends only on n , r , and r_0 , which are fixed, this implies that $f_k^{(n)}$ is Cauchy in $\overline{B}_{r_0}(z_0)$ if f_k is Cauchy on $\overline{B}_r(z_0)$. $\square$

The following is sort of a version of the maximum principle but for derivatives.

Lemma 3.7. Suppose f is holomorphic on $\overline{B}_r(z_0)$ and $0 < r_0 < r$. For all positive integers n ,

$$\sup_{B_{r_0}(z_0)} |f^{(n)}| \leq \frac{n!}{2\pi(r - r_0)^{n+1}} \int_{\partial B_r(z_0)} |f(z)| dz \leq \frac{n! \cdot r}{(r - r_0)^{n+1}} \sup_{\partial B_r(z_0)} |f|$$

Proof. By the inequality in the proof of Lemma 3.6 with f in place of f_k and the zero function in place of f_ℓ ,

$$|f^{(n)}(z)| \leq \frac{n!}{2\pi(r - r_0)^{n+1}} \int_0^{2\pi} |f(z_0 + re^{i\theta})| r d\theta \leq \frac{n! \cdot r}{(r - r_0)^{n+1}} \sup_{\partial B_r(z_0)} |f|$$

holds for all $z \in B_{r_0}(z_0)$, where ε is the supremum of $|f|$ and we have simply not yet estimated $|f|$ above by ε in the middle term. Then, the middle term is easily seen to be equal to the integral of $|f|$ over the boundary of $B_r(z_0)$, so taking the supremum over $z \in B_{r_0}(z_0)$ gives the inequality as stated in the lemma. $\square$

Theorem 3.8 (Liouville's theorem). If f is *entire*, meaning it is holomorphic on all of $\mathbb{C}$, and $|f|$ is bounded, then f is constant.

Proof. Let $M = \sup_{\mathbb{C}} |f| < \infty$. Lemma 3.7 says

$$|f'(z)| \leq \frac{R}{(R-|z|)^2} \cdot M$$

for all $R > |z|$, so taking R arbitrarily large implies that $f'(z) = 0$; thus $f'(z) = 0$ everywhere, meaning f is constant. $\square$

Theorem 3.9 (Fundamental theorem of algebra). Any nonconstant polynomial in $\mathbb{C}$ has a root in $\mathbb{C}$.

Proof. If $p(z) = a_0 + a_1z + \cdots + a_nz^n \in \mathbb{C}[z]$ has no roots, then $\frac{1}{p(z)}$ is an entire function. Since

$$\left| \frac{1}{p(z)} \right| = \frac{1}{|z|^n} \cdot \frac{1}{|a_n + a_{n-1}z^{-1} + \cdots + a_0z^{-n}|} < \frac{2}{|a_nz|^n}$$

holds if $|z|$ is large enough, $\frac{1}{p(z)}$ is bounded outside some ball $\overline{B}_R(z_0)$. On the other hand $\frac{1}{p(z)}$ is also bounded inside $\overline{B}_R(z_0)$ because $\overline{B}_R(z_0)$ is compact, so it is bounded on all of $\mathbb{C}$. Liouville's theorem then implies that $\frac{1}{p(z)}$, hence $p(z)$, must be constant if it has no roots. $\square$

3.3 Root counting

The Cauchy integral formula lets us *count the number of zeroes* of a holomorphic function!

Proposition 3.10 (Argument principle). Let Ω be a good domain and $f \in \mathcal{O}(\overline{\Omega})$. If $f|_{\partial\Omega}$ is nowhere zero, then f has finitely many, say N , zeroes in the interior of Ω , and

$$N = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)} dz.$$

Proof. The number of zeroes is finite because the set of zeroes of f is compact (because it is a closed subset of the compact set $\overline{\Omega}$) and discrete (since, by assumption, f is not everywhere zero, hence is not the zero function by Proposition 3.4), and compact, discrete sets must be finite.

If $z_1, \dots, z_N$ are the zeroes of f (where each zero is repeated in this list as many times as its multiplicity, e.g. $z_1 = z_2 = 0$ if $f(z) = z^2$), then $f(z) = (z - z_1) \cdots (z - z_N)g(z)$ where g is a nonzero holomorphic function on $\overline{\Omega}$; one can verify that g is indeed nonzero on $\overline{\Omega}$ by using analyticity. Then it is easy to compute

$$\frac{f'(z)}{f(z)} = \frac{1}{z - z_1} + \cdots + \frac{1}{z - z_N} + \frac{g'(z)}{g(z)}$$

using the product rule (the trick for computing this quickly is to formally differentiate the “function” $\log f = \log(z - z_1) + \cdots + \log(z - z_N) + \log g$). When integrating over $\partial\Omega$, the $\frac{1}{z - z_k}$ addends each become $2\pi i$ by the Cauchy integral formula, and since $\frac{g'(z)}{g(z)}$ is holomorphic on $\overline{\Omega}$, its integral is zero by the Cauchy integral theorem, so

$$\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)} dz = N.$$

Equivalently, one could apply the residue theorem (Theorem 3.20) to $\frac{f'(z)}{f(z)}$. $\square$

Geometrically, this says that the number of solutions to $f(z) = 0$ inside Ω is the winding number of $f(\partial\Omega)$ around 0 (as long as $f(\partial\Omega)$ does not pass through zero).

Corollary 3.11. Let $U \subseteq \mathbb{C}$ be open and connected and $f \in \mathcal{O}(U)$ nonconstant. For a given $z_0 \in U$, suppose that $f(z) - f(z_0) = (z - z_0)^n g(z)$ where $g \in \mathcal{O}(U)$ is nonzero at z_0 (n is called the *order* of the zero z_0 of f). Then, there exists $\delta > 0$ such that for all $a \in \mathbb{C}$ in $B_\delta(f(z_0))$, the equation $f(z) = a$ has n solutions in a neighborhood of z_0 .

Proof. Let $B_r(z_0)$ be a neighborhood of z_0 in which g is nonzero, so that

$$\frac{1}{2\pi i} \int_{\partial B_r(z_0)} \frac{f'(z)}{f(z) - f(z_0)} dz = n$$

by the argument principle. Then, for $a \in \mathbb{C} - f(\partial B_r(z_0))$, define

$$N(a) := \frac{1}{2\pi i} \int_{\partial B_r(z_0)} \frac{f'(z)}{f(z) - a} dz$$

to be the number of solutions to $f(z) = a$ inside $B_r(z_0)$. Then,

$$\begin{aligned} |N(z_0) - N(a)| &= \frac{1}{2\pi} \left| \int_{\partial B_r(z_0)} f'(z) \cdot \frac{a - f(z_0)}{(f(z) - a)(f(z) - f(z_0))} dz \right| \\ &= \frac{1}{2\pi} \left| \int_0^{2\pi} f'(z) \cdot \frac{a - f(z_0)}{(f(z) - a)(f(z) - f(z_0))} \cdot r i e^{i\theta} d\theta \right| \\ &\leq \frac{r |a - f(z_0)|}{2\pi} \int_0^{2\pi} \left| \frac{f'(z)}{(f(z) - a)(f(z) - f(z_0))} \right| d\theta \end{aligned}$$

Since $\partial B_r(z_0)$ is compact and $z_0 \notin f(\partial B_r(z_0))$, $|f(z) - f(z_0)|$ has a minimum value $M > 0$ on the circle. Supposing $|a - z_0| < \delta$ with $\delta < M$, we then also have $|f(z) - a| \geq M - \delta$ on the circle. Thus

$$\frac{r |a - f(z_0)|}{2\pi} \int_0^{2\pi} \left| \frac{f'(z)}{(f(z) - a)(f(z) - f(z_0))} \right| d\theta \leq \frac{r\delta}{M(M - \delta)} \int_0^{2\pi} |f'(z)| d\theta,$$

and the right-hand side can be made arbitrarily small by shrinking δ . Thus N is an integer-valued, continuous function on its domain, hence it is constant on the component of $\mathbb{C} - f(\partial B_r(z_0))$ containing z_0 . This component is open in $\mathbb{C}$ because it is an open subset of the open set $\mathbb{C} - f(\partial B_r(z_0))$ (noting that the image of the circle is compact hence closed), so one just needs to pick δ small enough that $B_\delta(f(z_0))$ is contained in this connected component. $\square$

Exercise 3.12 (Weak Rouché's theorem). Prove the following version of Rouché's theorem (Theorem 3.22) by appealing to the winding number interpretation of the argument principle: if f and g are holomorphic functions on good domain Ω such that $|f - g| < |g|$ on $\partial\Omega$, then f and g have the same number of zeroes inside Ω . The stronger (usual) version of this theorem, which we will see later, incorporates some information about the poles of f too. (Hint: you should be able to construct an explicit homotopy $H: (\mathbb{C} - \{0\}) \times I \rightarrow \mathbb{C} - \{0\}$ between $f(\partial\Omega)$ and $g(\partial\Omega)$.)

Exercise 3.13. Use Rouché's theorem to prove Corollary 3.11.

Two super important corollaries:

Theorem 3.14 (Open mapping theorem). Let $U \subseteq \mathbb{C}$ be open and connected, and $f \in \mathcal{O}(U)$ nonconstant. Then f is an open map.

Proof. This follows immediately from Corollary 3.11, since the corollary implies that if a is contained in $f(U)$ then so is some open ball around a . $\square$

Theorem 3.15 (Maximum modulus principle). Let Ω be a good domain and $f \in \mathcal{O}(\overline{\Omega})$ a holomorphic function. For every z_0 in the interior of Ω ,

$$|f(z_0)| < \sup_{z \in \partial\Omega} |f(z)|$$

(note the strict inequality).

Proof. The open mapping theorem tells us that f sends the interior of $\overline{\Omega}$ to the interior of $f(\overline{\Omega})$. Since $\overline{\Omega}$ is compact, $f(\overline{\Omega})$ is too, so there is a point in $f(\overline{\Omega})$ maximizing the magnitude over all points in $f(\overline{\Omega})$; any such point necessarily cannot be in the interior of $f(\overline{\Omega})$, so neither can its preimage under f . Thus any point maximizing the magnitude of f over $\overline{\Omega}$ cannot be in the interior of Ω . $\square$

3.4 Poles

Let $U \subseteq \mathbb{C}$ be open and $z_0 \in U$. If $f \in \mathcal{O}(U - \{z_0\})$, we say that z_0 is an *isolated singularity* of f . There are three possibilities for the behavior of f around z_0 .

- (1) If f extends to a holomorphic function on U , then we call z_0 a *removable singularity* of f . Such points are characterized by $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$, in which case we can define $\tilde{f} \in \mathcal{O}(U)$ by taking $\tilde{f} \equiv f$ on $U - \{z_0\}$ and

$$\tilde{f}(z_0) = \frac{1}{2\pi i} \int_{\partial B_\varepsilon(z_0)} \frac{f(\zeta)}{\zeta - z_0} d\zeta$$

(which is of course preordained by the Cauchy integral formula).

Exercise 3.16. If f is a removable singularity (in the sense of the limit definition) show that the function $g: U \rightarrow \mathbb{C}$ defined by $g(z_0) = 0$ and $g(z) = (z - z_0)^2 f(z)$ for $z \in U - \{z_0\}$ is holomorphic on U , and that z_0 is a zero of g with order at least 2. Conclude that f admits an extension to U .

- (2) It could also happen that there is a positive integer n such that $g(z) = (z - z_0)^n f(z)$ extends to a nonzero holomorphic function in a neighborhood of z_0 . In this case, z_0 is a *pole* of f with *order* n . Recall that, dually, z_0 is a zero of f with order n if $f(z) = (z - z_0)^n h(z)$ for some holomorphic h that is nonzero in a neighborhood of z_0 — poles and zeroes thus behave “opposite” to each other (cf. Proposition 3.21 and Theorem 3.22). A holomorphic function whose isolated singularities are all poles (or removable, but we can just remove those) is called *meromorphic*; these functions can be thought of as holomorphic functions to the Riemann sphere (which is another reason why zeroes and poles behave dually to one another). We will say more about meromorphic functions momentarily.
- (3) Finally, in the case where z_0 is neither removable nor a pole, we say that z_0 is an *essential singularity* — these have incredibly interesting behavior.

Here is one example of the crazy behavior of essential singularities. (In my head, the situation is just that cases (1) and (2) are so rigid and well-behaved that deviating at all from those scenarios causes things to explode. Maybe the right adjective is “brittle”.)

Proposition 3.17 (Casorati-Weierstrass). If z_0 is an essential singularity of f , then the image of any neighborhood of z_0 under f is dense in $\mathbb{C}$.

Proof.

$\square$

For example, $f(z) = e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!z^n}$ has an essential singularity at zero.

Now, we will talk about meromorphic functions. The most important definition relating to meromorphic functions is the idea of a residue.

todo; idea is just that if it isn't dense then we get a rem sing or pole

Definition 3.18 (Residue) — If f is holomorphic in a punctured neighborhood of $a \in \mathbb{C}$ (so not necessarily holomorphic at a itself), then the *residue* of f at a is

$$\text{Res}_a(f) = \frac{1}{2\pi i} \int_{|z-a|=\varepsilon} f(z) dz$$

where ε is small.

Of course, residues are only interesting at singularities, since if f is holomorphic at a then the residue is zero by Cauchy's integral theorem (Theorem 3.1).

Proposition 3.19. Suppose a is a pole of f with order n . Then

$$\text{Res}_a(f) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} ((z-a)^n f(z)) \Big|_{z=a}.$$

Proof. By definition of pole, $(z-a)^n f(z)$ extends to a nonzero holomorphic function in a neighborhood of a , so there exist complex numbers c_k such that

$$(z-a)^n f(z) = c_0 + c_1(z-a) + c_2(z-a)^2 + \cdots,$$

with $c_0 \neq 0$, in a neighborhood of a . This means that

$$f(z) = \frac{c_0}{(z-a)^n} + \frac{c_1}{(z-a)^{n-1}} + \cdots + \frac{c_{n-1}}{z-a} + c_n + c_{n+1}(z-a) + \cdots$$

in a punctured neighborhood of a . Integrating over a sufficiently small circle centered at a kills every addend except $\frac{c_{n-1}}{z-a}$ (where we can integrate termwise because power series converge locally uniformly), so that

$$\text{Res}_a(f) = \frac{1}{2\pi i} \int_{\partial B_\varepsilon(a)} f(z) dz = \frac{1}{2\pi i} \int_{\partial B_\varepsilon(a)} \frac{c_{n-1}}{z-a} dz = c_{n-1}.$$

On the other hand we can compute c_{n-1} in the standard way by differentiating the power series for $(z-a)^n f(z)$ above $n-1$ times, dividing by $(n-1)!$, and evaluating at a . $\square$

In the case where $n = 1$, the formula above involves the zeroth derivative of $(z-a)f(z)$, which is just the value of (the extension to a neighborhood of a of) $(z-a)f(z)$ at a , i.e.

$$\text{Res}_a(f) = \lim_{z \rightarrow a} (z-a)f(z).$$

Such a pole is called a *simple pole* of f .

The reason that we care to define residues is that the integral of a meromorphic function is dependent only on the residues:

Theorem 3.20 (Residue theorem). Let f be a meromorphic function defined on good domain Ω . Then, f has finitely many poles $a_1, \dots, a_N$, and

$$\frac{1}{2\pi i} \int_{\partial\Omega} f(z) dz = \sum_{k=1}^N \text{Res}_{a_k}(f).$$

Proof. Since the poles of f are the zeroes of $1/f$ (or rather, we can extend $1/f$ to a meromorphic function whose zeroes are the poles of f), the set of poles is closed, so it is compact and discrete, hence finite.

In particular, there exists $\varepsilon > 0$ such that the ε -balls around each a_k are pairwise disjoint, meaning f is holomorphic on $\tilde{\Omega} = \Omega - \bigcup_{k=1}^N B_\varepsilon(a_k)$, so that

$$0 = \int_{\partial\tilde{\Omega}} f(z) dz = \int_{\partial\Omega} f(z) dz - \sum_{k=1}^N \int_{\partial B_\varepsilon(a_k)} f(z) dz = \int_{\partial\Omega} f(z) dz - \sum_{k=1}^N \text{Res}_{a_k}(f). \quad \square$$

One way to think about the argument principle (Proposition 3.10) is that holomorphic functions look like polynomials of degree n in a neighborhood of a zero of order n , which we can glean from the power series, as higher-order terms are small in comparison to the lowest one. This explains why the winding number of $f(\partial B_\varepsilon(z_0))$ around 0 is n if ε is small enough and z_0 is an isolated zero of order n . (Green's theorem lets us patch this local interpretation into a more global integral, in the same way that the residues of a meromorphic function determine its integral.)

In analogy, a meromorphic function f looks like $(z - a_0)^{-n}$ around a pole a_0 of order n , which implies that the winding number of $f(B_\varepsilon(z_0))$ for small ε should be $-n$ (since for example $1/z$ reverses the orientation of the circle). Thus, the poles are “negative zeroes” for the purposes of counting facts like the argument principle:

Proposition 3.21 (Argument principle for poles). Let f be meromorphic on a good domain Ω with no zeroes or poles on $\partial\Omega$. Then

$$\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)} dz = N(f, \Omega) - P(f, \Omega)$$

where $N(f, \Omega)$ and $P(f, \Omega)$ are respectively the number of zeroes and poles of f in Ω .

Note that the integral on the left gets negated when replacing f with $1/f$, which makes sense because taking the reciprocal swaps zeroes and poles.

Proof of Proposition 3.21. Note that $n = N(f, \Omega)$ and $p = P(f, \Omega)$ are both finite. We do a similar argument to the proof of Proposition 3.10: if $a_1, \dots, a_n$ are the zeroes of f and $b_1, \dots, b_p$ are the poles, both lists with repeats according to multiplicity, then

$$f(z) = \frac{(z - a_1) \cdots (z - a_n)}{(z - b_1) \cdots (z - b_p)} g(z)$$

where g extends to a nowhere zero holomorphic function on Ω . Then, by the product rule or logarithmic differentiation,

$$\frac{f'(z)}{f(z)} = \sum_{k=1}^n \frac{1}{z - a_k} - \sum_{k=1}^p \frac{1}{z - b_k} + \frac{g'(z)}{g(z)},$$

and then taking the integral gives

$$\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)} dz = \sum_{k=1}^n 1 - \sum_{k=1}^p 1 + 0 = n - p$$

by the Cauchy integral formula for the $\frac{1}{z-w}$ summands and the Cauchy integral theorem for $\frac{g'(z)}{g(z)}$. $\square$

Recall Exercise 3.12 tells us how if two holomorphic functions f and g send $\partial\Omega$ sufficiently close to one another (in the sense of the given inequality), then f and g have the same number of zeroes in Ω . Since that version of Rouché's theorem relied on only the argument principle, there is a version for meromorphic functions:

Theorem 3.22 (Rouché's theorem). Let f and g be meromorphic functions on Ω . If $|f - g| < |f|$ on $\partial\Omega$, then $N(f, \Omega) - P(f, \Omega) = N(g, \Omega) - P(g, \Omega)$.

The homotopy idea in Exercise 3.12 can also be used here, but here is another proof.

Proof of Rouché's theorem. The inequality says that $|\frac{g}{f} - 1| < 1$ on $\partial\Omega$, so $F = \frac{g}{f}$ maps $\partial\Omega$ into the open disk $B_1(1)$. Let

$$n(a) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{F'(z)}{F(z) - a} dz$$

be the number of times F attains the value $a \in \mathbb{C} - F(\partial\Omega)$ in Ω , which is constant on each component of $\mathbb{C} - F(\partial\Omega)$. Obviously $n(a)$ is zero on the unbounded component (since e.g. $|\frac{g}{f}|$ is bounded on $\partial\Omega$), and 0 is contained in the unbounded component, so number of zeroes of F is

$$\begin{aligned} 0 = n(0) &= \int_{\partial\Omega} \frac{F'(z)}{F(z)} dz = N\left(\frac{g}{f}, \Omega\right) - P\left(\frac{g}{f}, \Omega\right) \\ &= (N(g, \Omega) + P(f, \Omega)) - (P(g, \Omega) + N(f, \Omega)), \end{aligned}$$

where we recall that taking the reciprocal swaps poles and zeroes. □

laurent series

3.5 Schwarz lemma

The maximum principle (and the open mapping theorem) are very strong structural statements, and with some cleverness we can extract a lot of juice out of them.

Theorem 3.23 (Schwarz lemma). Let Δ be the open unit disk and $f \in \mathcal{O}(\Delta)$ such that

- (1) $|f| < 1$ on Δ ; and
- (2) $f(0) = 0$.

Then $|f(z)| < |z|$ on all of Δ and $|f'(0)| \leq 1$. Furthermore, if $|f(z)| = |z|$ for any nonzero $z \in \Delta - \{0\}$ or $|f'(0)| = 1$, then $f(z) = cz$ for some $c \in S^1$.

Proof. Let $g(z) = \frac{f(z)}{z}$. Since $f(0) = 0$, f has a power series expansion $f(z) = c_1 z + c_2 z^2 + \dots$ that converges on all of Δ (examine the proof of holomorphicity implying analyticity, Theorem 3.3). Define $g(z) = c_1 + c_2 z + \dots$ to be the extension of $\frac{f(z)}{z}$ to Δ by setting $g(0) = f'(0)$; this function is holomorphic because we have defined it via a power series.

Given $z \in \Delta$, we have by the maximum principle (Theorem 3.15) that

$$|g(z)| \leq \sup_{w \in B_r(0)} \frac{1}{|w|} \leq \frac{1}{r}$$

for any r satisfying $|z| \leq r < 1$, so taking r arbitrarily close to 1 tells us that $|g| \leq 1$, which is equivalent to the inequalities that we want to prove. The two equality conditions are both equivalent to saying that $|g(z)| = 1$ for some $z \in \Delta$; this would imply that z is a local maximum for g in the interior of $B_r(0)$ for any r with $|z| < r < 1$, which implies that g must be constant on $B_r(0)$. If this is the case then $g(0) = c_1$ has magnitude 1 and $f(z) = c_1 z$ for all $z \in \Delta$, as desired. □

Corollary 3.24. The automorphisms of the unit disk (i.e. holomorphic bijections $\Delta \rightarrow \Delta$ with holomorphic inverse) are given by the fractional linear transformations $\varphi(z) = c \frac{a-z}{1-\bar{a}z}$ for $|a| < 1$, $|c| = 1$.

Proof. Let $a \in \Delta$ be the point with $f(a) = 0$, $\varphi(z) = \frac{a-z}{1-\bar{a}z}$, and $g(z) = f(\varphi(z))$. Then, g and g^{-1} are automorphisms of Δ and satisfy the hypotheses of the Schwarz lemma, so $|g'(0)| \leq 1$ and $|(g^{-1})'(0)| \leq 1$. The chain rule tells us that $g'(0)(g^{-1})'(0) = 1$, so the inequalities above are equalities, meaning $g(z) = cz$ for some $|c| = 1$. □

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define holomorphic map between riemann surfaces	5
idk difficulty of pf	5
give another example of riemann surface using log	5
write down power series facts	5
induced maps on tangent/cotangent spaces via a smooth map $\mathbb{R}^m \rightarrow \mathbb{R}^n$	5
good domains, orientation, Green's theorem	5
write down important results from HWs	6
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